

Sanctions Risk and Dollar Convenience

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Abstract

Financial sanctions derive power from the dollar network and can erode that network when reserve managers learn that dollar access is state contingent. I study the policy rule that joins these forces. In a local reserve-demand model, expected inaccessibility is a tax on convenience and network complementarity amplifies the aggregate response. Heterogeneous portfolio costs prevent the network term from canceling in raw country comparisons, so the empirical object is a conditional average direct semi-elasticity. A fixed-belief discretion benchmark delivers maximal sanctions, while public belief updating makes reputation a state variable and turns that result into a punishment allocation rather than a general law of discretion. Small sanctions reassessments are evaluated in the local model. Routine use is evaluated with bounded convenience benefits and convex portfolio costs matched to the same local equilibrium. The quantitative results are scenario thresholds. At the baseline accounting share $\zeta_H = 0.50$, full use ceases to dominate restraint when expected breadth reaches 5.5 percent. The threshold moves to 2.8 percent when crisis liquidity doubles the actuarial loss and to 1.4 percent when it quadruples. The contribution is the interaction between sanctions rules and an endogenous reserve-currency network, not the sanctions wedge by itself.

JEL codes: F31, F33, F51, E42, G15.

Keywords: financial sanctions, reserve currencies, convenience yield, network externalities, time inconsistency.

1 Introduction

In the last days of February 2022 the United States and its allies froze roughly \$300 billion of assets belonging to the central bank of Russia: about half of that country's reserves, and the first such action ever taken against a central bank of the Group of Twenty (Ribakova et al., 2025). The freeze worked through accounts and custodians. It worked because Russia, like nearly every country, held its safe assets inside the currencies and custodial systems of a small group of Western jurisdictions. The power of the instrument comes from the size of the dollar network. The question of this paper is what exercising that power costs its owner.

The cost, if there is one, operates through a belief. A reserve manager outside the United States alliance network watched February 2022 and learned that the safety of dollar assets is state contingent: safe in most states of the world, inaccessible in the state where his government

collides with Washington. Treasury yields and default risk stayed fixed. What changed was the probability each holder attaches to being able to use them when they are most needed. That probability times the expected haircut is a wedge, denominated in the same units as the convenience yield that makes reserve assets worth holding at a premium in the first place (Krishnamurthy and Vissing-Jorgensen, 2012). This paper prices the wedge, traces it through the network externality that sustains a dominant currency, and asks how a hegemon that understands the whole chain should use, or decline to use, the financial weapon.

The local benchmark has a sharp decomposition. Countries choose the dollar share of reserves by balancing convenience against portfolio concentration. Expected inaccessibility subtracts from convenience in the same units. With common portfolio curvature and common network exposure, the aligned-exposed difference identifies the sanctions wedge divided by marginal portfolio cost. Aggregate demand responds by the multiplier $m = 1/(1 - \ell_1/\kappa)$. This cancellation is a property of the homogeneous benchmark, not an identification theorem for raw country data. With $\kappa_i = \exp(X_i'\beta_\kappa + \eta_i)$ and country-specific network loadings, the local estimand is an average direct response conditional on X_i , and the aggregate multiplier depends on the distribution of those loadings.

Adjustment costs turn a permanent reassessment into slow portfolio drift. The local saddle path separates the exposed-aligned gap from the aggregate network response. It supplies testable timing restrictions, but the three-year half-life remains an external scenario until estimated from portfolio data. Aggregate COFER cannot identify it by itself because valuation, reserve accumulation, and asset returns move the same series.

The hegemon's problem contains a commitment force. Under fixed beliefs, portfolios are predetermined at the crisis date and the ex post choice is maximal. This is a no-reputation benchmark. When reserve managers update a public reputation state after observed sanctions, current intensity changes future demand. The Ramsey rule internalizes that effect. A trigger strategy provides a transparent repeated-game implementation, but maximal sanctions is then the off-path punishment rather than an unconditional prediction.

The quantitative section reports a sanctions-use frontier. The issuer captures only a share ζ_H of reserve convenience, and reserve losses receive a crisis-liquidity multiplier χ_i . Geopolitical stakes remain a policy input. The calculation therefore asks which geopolitical benefit justifies a proposed breadth of use. At breadth 0.02 the baseline benefit threshold is 71.4 billion dollars per crisis. At breadth 0.10 it is 363.9 billion. These are inversion results, not welfare estimates.

Bianchi and Sosa-Padilla (2025) establish that anticipated financial sanctions can reduce dollar convenience and dollar holdings. The sanctions wedge is therefore not the novelty claimed here. The contribution is to close the policy-rule loop through an endogenous currency network and to distinguish the local portfolio response from a bounded global scenario. The network block builds on Gopinath and Stein (2021), Farhi and Maggiori (2018), and He et al. (2019). Convenience-yield measures provide units, while public TIC holdings and aggregate COFER discipline different objects.

The linear economy governs only small reassessments around the observed reserve equilibrium.

The global routine-use scenario uses bounded convenience and stronger portfolio curvature. The target country's response is outside the model. Geopolitical benefits are not inferred from military spending or fiscal aid. They enter only through reported threshold functions.

2 Reserves, Convenience, and the Sanctions Wedge

The linear-quadratic structure is the local economy in which the results are derived. Propositions 1 through 5 are exact inside that economy. The sign results use the slope of convenience in network size. The magnitudes use the linear form and the calibration in Section 5. Appendix C extends the same logic to corners.

2.1 Environment

There is a continuum of countries $i \in [0, 1]$ and a dominant-currency issuer, the hegemon. Each period, country i allocates its reserves between dollar assets, in share $a_i \in [0, 1]$, and an outside reserve asset: gold, or any other store of value beyond the hegemon's legal reach. Dollar assets pay a financial return spread $\bar{r}$ over the outside asset (zero in the baseline calibration) and deliver liquidity services

$$\ell(N) = \ell_0 + \ell_1 N, \quad \ell_1 \geq 0, \quad N = \int_0^1 a_i di, \quad (1)$$

where N is the aggregate dollar share: the size of the dollar network. The dependence of ℓ on N is the reduced form of the complementarities that the dominant-currency literature microfounds, deeper markets and invoicing that matches the reserve asset (Gopinath and Stein, 2021; Farhi and Maggiori, 2018; He et al., 2019). I keep it linear for two reasons. The results are then exact, and measurement disciplines a slope, the local semi-elasticity delivered by the empirical designs of Section 6. The multiplier below is a constant rather than a state variable, while the global phenomena generated by curvature, tipping and multiplicity, enter through the corner regime of Appendix C. Around the observed equilibrium, linearity is the first-order approximation to any smooth $\ell(N)$, so the local comparative statics are general. The routine-use calibration of Section 5 is solved with the corner-safe algorithm described there.

Sanctions enter as an expected loss of liquidity services. Country i has exposure $z_i \in [0, 1]$. A mass μ of reserves is held by exposed countries and the rest is held inside the hegemon's security orbit. With probability ω each period a geopolitical crisis arrives. The hegemon's rule assigns intensity $\bar{s} \in [0, 1]$. Conditional on a crisis, country i is targeted with probability p_i and loses access to a fraction ϕ_i of its dollar reserves. The loss is most costly when reserves are needed. Let

$$\hat{\pi}_i = \omega \bar{s} p_i \phi_i \chi_i, \quad \chi_i \equiv \frac{u'_i(L_i^{\text{crisis}})}{u'_i(L_i^{\text{normal}})} \geq 1, \quad (2)$$

where χ_i is the crisis-liquidity multiplier. The object $z_i \hat{\pi}_i$ is the *sanctions wedge*. It is a state-contingent haircut on the liquidity value of dollar reserves, measured in annual return units. This is the anticipation channel in Bianchi and Sosa-Padilla (2025). Reserve demand prices the

product in (2). Its decomposition matters for policy counterfactuals and cannot be recovered from reserve quantities alone.

Country i 's per-period objective over its reserve share is quadratic:

$$u_i(a, a_{-1}) = a(\bar{r} + \lambda_i \ell(N) - z_i \hat{\pi}_i) - \frac{\kappa_i}{2} a^2 - \frac{\nu_i}{2} (a - a_{-1})^2. \quad (3)$$

The parameter λ_i measures exposure to the dollar network. The parameter κ_i prices concentration costs arising from currency mismatch, trade invoicing, and political diversification. The parameter ν_i prices gradual portfolio adjustment. The homogeneous benchmark sets $\lambda_i = 1$, $\kappa_i = \kappa$, $\nu_i = \nu$, and $\hat{\pi}_i = \hat{\pi}$ for exposed countries. Set $\nu = 0$ for now. Section 3 restores it.

2.2 Pricing the weapon

In the homogeneous benchmark, the interior first-order condition gives

$$a_i = \frac{\bar{r} + \ell(N) - z_i \hat{\pi}}{\kappa}. \quad (4)$$

Proposition 1 (The sanctions wedge is priced like a tax on convenience). *At any interior equilibrium: (i) the dollar share of an exposed country falls one-for-one with $\hat{\pi}$ scaled by $1/\kappa$, holding N fixed. (ii) The equilibrium difference between aligned and exposed portfolios is*

$$a_A - a_E = \frac{\hat{\pi}}{\kappa},$$

independent of the network term $\ell(N)$. (iii) The effective convenience yield earned by an exposed country is $\ell(N) - \hat{\pi}$: sanctions risk and liquidity services are the same object with opposite signs.

Convenience yields are measured in basis points (Krishnamurthy and Vissing-Jorgensen, 2012; Du et al., 2018). Equation (2) places accessibility risk in the same units. With $\chi = 1$, a freeze that arrives once every twenty years, strikes a given country with probability 0.02 conditional on a crisis, and blocks one half of its portfolio creates a five-basis-point wedge. A crisis multiplier of two doubles that wedge without changing the actuarial probability.

2.3 The network multiplier

Aggregating (4) over countries and solving for N :

$$N = \frac{\bar{r} + \ell_0 - \mu \hat{\pi}}{\kappa - \ell_1}, \quad \text{so} \quad \frac{dN}{d\hat{\pi}} = -\frac{\mu}{\kappa} m, \quad m \equiv \frac{1}{1 - \ell_1/\kappa}. \quad (5)$$

Proposition 2 (Multiplier and uniqueness). *If $\ell_1 < \kappa$ and the solution is interior, the equilibrium exists and is unique. A sanctions-risk shock that lowers exposed demand directly by $\mu \hat{\pi}/\kappa$ lowers the aggregate network by m times that amount, where $m > 1$ whenever $\ell_1 > 0$. If $\ell_1 \geq \kappa$, the contraction argument fails. Interior equilibria can be unstable and the constrained economy can display corner equilibria or multiplicity.*

The quantitative local economy satisfies $\ell_1 < \kappa$. Appendix C treats constrained choices. The condition is sufficient for uniqueness and cannot be read backward as a theorem of multiplicity.

Corollary 1 (Micro and macro). *In the homogeneous benchmark the cross-sectional difference $a_A - a_E = \hat{\pi}/\kappa$ differences out the common network term. The aggregate response in (5) is m times the direct response. Identification of m requires the cross-sectional response together with an aggregate quantity or convenience-yield moment.*

The homogeneous cancellation is useful because it states what must be controlled when countries differ. It is not an interpretation of raw country comparisons.

2.4 Heterogeneous reserve managers

Set $\nu_i = 0$ and retain interior choices. The policy is

$$a_i = \frac{\bar{r} + \lambda_i \ell(N) - z_i \hat{\pi}_i}{\kappa_i}. \quad (6)$$

Define $A_\lambda = \int_0^1 \lambda_i / \kappa_i di$ and, for a common proportional change in sanction intensity, $A_z = \int_0^1 z_i \omega p_i \phi_i \chi_i / \kappa_i di$.

Proposition 3 (Heterogeneous local response). *If $1 - \ell_1 A_\lambda > 0$, the local equilibrium is unique and*

$$\frac{dN}{d\bar{s}} = -\frac{A_z}{1 - \ell_1 A_\lambda}. \quad (7)$$

The direct semi-elasticity of country i is $-z_i \omega p_i \phi_i \chi_i / \kappa_i$. A group comparison also contains differences in λ_i / κ_i multiplied by the equilibrium network response.

Equation (7) is the empirical object. Countries with the same geopolitical exposure can respond differently because crisis liquidity, portfolio curvature, and reliance on dollar invoicing differ. Aligned countries respond through $\lambda_i \ell_1 dN$. Their movement is evidence consistent with a network channel, but it identifies ℓ_1 only after the distribution of λ_i / κ_i and common shocks are disciplined.

The dynamic model supplies the pace, and the pace carries the identification. That is the next section.

3 Dynamics: Reassessment and Slow Erosion

Restore the adjustment cost $\nu > 0$ and let countries discount at β_c . The first-order condition of the forward-looking problem is the Euler equation

$$\bar{r} + \ell(N_t) - z_i \hat{\pi} - \kappa a_{i,t} - \nu(a_{i,t} - a_{i,t-1}) + \beta_c \nu (\mathbb{E}_t a_{i,t+1} - a_{i,t}) = 0. \quad (8)$$

Steady states are unchanged: the ν terms cancel, so (4)–(5) still describe where the system settles. What ν governs is the road. Aggregating (8) gives a linear second-order difference equation in N_t ,

and differencing across types gives an autonomous equation for the gap $d_t = a_{E,t} - a_{A,t}$. Both solve by factoring a quadratic (Appendix B).

Proposition 4 (Saddle path). *Suppose $\ell_1 < \kappa$. After a permanent change in $\hat{\pi}$, the unique bounded perfect-foresight path satisfies*

$$N_t - N_\infty = \lambda_N (N_{t-1} - N_\infty), \quad d_t - d_\infty = \lambda_d (d_{t-1} - d_\infty),$$

where $\lambda_N \in (0, 1)$ is the stable root of $\beta_c \nu \lambda^2 - (\kappa + (1 + \beta_c) \nu - \ell_1) \lambda + \nu = 0$ and λ_d solves the same equation with $\ell_1 = 0$. The aggregate is more persistent than the cross-section, $\lambda_N > \lambda_d$, and λ_N is increasing in ℓ_1 : complementarity adds inertia as well as amplification.

The proposition concerns a permanent reassessment of $\hat{\pi}$. Under that maintained shock, reserve shares drift toward a lower plateau. A transitory risk shock instead generates reversion. The distinction must be estimated rather than inferred from the slow decline in aggregate COFER (Arslanalp et al., 2022). The ranking $\lambda_N > \lambda_d$ supplies a separate timing restriction. The exposed-aligned gap approaches its plateau faster than the aggregate share, so the country panel and aggregate series must agree on different roots of the same adjustment system.

Who, then, should fire such a weapon, and how often? The demand side is now complete, and the question moves to the hegemon.

4 The Hegemon and the Weaponization Bias

4.1 The hegemon's objective

Foreign official investors accept a return lower by $\ell(N)$ than their outside opportunity. Only a share of that saving reaches the issuer. Intermediation rents, regulatory costs, and foreign holdings outside direct sovereign liabilities absorb the remainder. Let $\zeta_H \in [0, 1]$ denote the issuer's capture rate. With W denoting world reserves, the privilege flow is

$$\Omega(N) = \zeta_H \ell(N) N W, \tag{9}$$

which is increasing in N in the maintained local region. Geopolitical crises arrive with probability ω per period. In a crisis the hegemon chooses sanction intensity $s \in [0, 1]$ against the target and collects

$$G(s) = \gamma \left(s - \frac{1}{2} s^2 \right), \tag{10}$$

where γ is a policy input. The functional form normalizes the ex post optimum to $s = 1$ when portfolio beliefs are fixed.

The timing within a period is the natural one and does all the work: reserve portfolios are set first, on the basis of the anticipated rule $\bar{s}$ through the wedge (2), and the crisis, if any, is realized after. Today's sanction moves tomorrow's portfolios through what it teaches about the rule.

4.2 Discretion and the rule

Consider the benchmark in which the hegemon chooses s crisis by crisis and a single action cannot revise reserve managers' beliefs.

Assumption 1 (Fixed beliefs at the crisis date). *At the date of a crisis, portfolios and the anticipated future rule $\bar{s}$ are predetermined. The current choice of s does not move either object within the period.*

Under Assumption 1, the ex post problem is $\max_s G(s)$ and its solution is $s_D = 1$. This is a fixed-belief punishment allocation, not a general theorem that discretion always produces maximal sanctions. Once public actions revise beliefs, reputation is a state and the continuation value enters the crisis decision.

The Ramsey hegemon chooses the rule itself, internalizing $N(\bar{s})$ from (5):

$$\max_{\bar{s} \in [0,1]} V(\bar{s}) = \Omega(N(\bar{s})) + \omega G(\bar{s}). \quad (11)$$

Proposition 5 (Weaponization bias). *Let $c \equiv \mu p \phi \chi / (\kappa - \ell_1)$ and $\bar{N}_0 = (\bar{r} + \ell_0) / (\kappa - \ell_1)$. Suppose the allocation is interior and $\gamma > 2\zeta_H \ell_1 \omega c^2 W$. The Ramsey rule is*

$$s_R = \frac{\gamma - \zeta_H c W (\ell_0 + 2\ell_1 \bar{N}_0)}{\gamma - 2\zeta_H \ell_1 \omega c^2 W} \quad (12)$$

clamped to $[0, 1]$. If $c > 0$ and the marginal captured privilege is positive at $N(1)$, then $s_R < s_D = 1$. The gap is the cost of taking portfolio beliefs as fixed at the crisis date.

The rule becomes more restrained when a given action destroys more captured privilege. This occurs when target breadth, freeze severity, crisis liquidity, issuer pass-through, or the network response rises. As $\gamma \rightarrow \infty$, the Ramsey and fixed-belief choices converge.

4.3 Reputation

Let r_t denote the public reputation state and let x_t collect geopolitical fundamentals. Beliefs evolve according to a public transition rule $r_{t+1} = \mathcal{B}(r_t, s_t, x_t)$. The hegemon's recursive problem is

$$\mathcal{V}(r, N, x) = \max_{s \in [0,1]} \{ \omega G(s, x) + \Omega(N) + \beta_H \mathbb{E}[\mathcal{V}(r', N', x') \mid r, N, x, s] \}. \quad (13)$$

This formulation makes clear why the maximal choice is limited to the fixed-belief benchmark. For a quantitative bound, specialize to a public trigger rule. Reserve managers price $\bar{s} = s_R$ while the rule is respected and price $\bar{s} = 1$ after a public deviation. Deviating gains $G(1) - G(s_R)$ once. Conformity preserves the continuation flow gap.

Proposition 6 (Reputation). *In the interior regime, the trigger equilibrium sustains the Ramsey rule if and only if the hegemon's discount factor β_H satisfies*

$$\frac{\beta_H}{1 - \beta_H} \geq \frac{\gamma}{\omega \gamma - 2\zeta_H \ell_1 \omega^2 c^2 W}, \quad (14)$$

Table 1: Calibration (annual).

Parameter		Value	Category	Basis
Convenience yield, pre-2022	$\ell(N_0)$	73 bp	external estimate	AAA–Treasury spread (KVJ)
Dollar share, pre-2022	N_0	0.59	observed aggregate	COFER
World reserves	W	\$12 tn	observed aggregate	COFER
Freeze severity	ϕ	0.5 {0.25, 0.75}	observed episode	Russia 2022 asset freeze
Diversification cost	κ	124 bp	model implied	$\kappa = \ell(N_0)/N_0$
Network multiplier	m	1.5 {1, 2.5}	scenario	identified by joint moments
Adjustment cost	ν	0.133	scenario implied	3-year aggregate half-life
Country discount factor	β_c	0.96	external convention	annual value
Exposed reserve share	μ	0.5 {0.3, 0.7}	scenario	coalition classification
Crisis arrival	ω	0.05	scenario	one event per two decades
Target breadth	p	0.02	scenario	threshold input
Crisis liquidity	χ	1 {1, 2, 4}	scenario	one is actuarial lower bound
Issuer capture	ζ_H	0.50 {0.25, 1}	scenario	privilege pass-through
Geopolitical stakes	γ	\$200 bn {50–800}	policy input	threshold reported below

Notes: KVJ is Krishnamurthy and Vissing-Jorgensen (2012). COFER is public only in aggregate. Braces give reported scenario ranges. No scenario input is described as an estimate.

with c as in Proposition 5. The condition is specific to this public trigger strategy and the interior local economy.

At the baseline scenario the trigger threshold corresponds to an annual discount rate of 5.0 percent. This calculation does not establish that observed policy follows a trigger equilibrium. It reports the patience required by one transparent enforcement technology. The global routine-use scenario is evaluated numerically because its continuation payoff is not quadratic.

5 Quantitative Analysis

5.1 Calibration

Table 1 separates observations and external estimates from model-implied objects and scenario inputs. This distinction governs the interpretation of every number below.

The observed pair $\ell(N_0) = 73$ basis points and $N_0 = 0.59$ implies $\kappa = 124$ basis points under the normalization $\bar{r} = 0$. This is a model-implied curvature, not a direct portfolio estimate. The baseline wedge is 5.0 basis points when $p = 0.02$ and $\chi = 1$. The remaining geopolitical quantities are inputs to threshold functions. They are not estimates of an observed policy objective.

5.2 Local and global economies

The linear model is used for the 2022 reassessment and the proofs in Sections 2–4. Routine use moves reserve shares too far for a local approximation. The global scenario replaces linear convenience with

$$\ell^g(N) = \ell_{\min} + \frac{\ell_{\max} - \ell_{\min}}{1 + \exp\{-b(N - \bar{N})\}} \quad (15)$$

Table 2: Steady states: the price of the weapon by regime of use.

	Pre-2022 ($\hat{\pi} = 0$)	Single use ($p = 0.02$)	Soft restrictions ($\phi = 0.15$)	Routine use ($p = 25\%$)
Dollar share N (percent)	59	56.0	58.1	23.2
exposed a_E	59	53.9	57.5	0.0
aligned a_A	59	58.0	58.7	46.5
Convenience yield (bp)	73	71.7	72.6	58
Captured privilege (\$bn/yr)	25.8	24.1	25.3	8.1

Notes: the single-use and soft-restriction columns use the local economy. The routine-use column uses the matched global economy. Captured privilege sets $\zeta_H = 0.50$. All geopolitical parameters are scenario inputs.

and replaces quadratic portfolio cost by

$$C'(a) = qa + \tau a^3. \quad (16)$$

The code sets $\ell_{\min} = 0.001$, $\ell_{\max} = 0.012$, and $\tau = 0.002$. The remaining coefficients solve

$$\ell^g(N_0) = \ell(N_0), \quad \ell^{g'}(N_0) = \ell_1, \quad C''(N_0) = \kappa, \quad (17)$$

with $\bar{r} + \ell^g(N_0) - C'(N_0) = 0$. The global economy shares the observed equilibrium and its local derivatives with the linear economy. It differs only away from that point.

Proposition 7 (Global scenario equilibrium). *Suppose $C_i''(a) > 0$ on $[0, 1]$. If the slope of the aggregate best response satisfies*

$$\sup_{N \in [0, 1]} \left\{ \ell^{g'}(N) \int_{\mathcal{I}(N)} \frac{\lambda_i}{C_i''(a_i(N))} di \right\} < 1, \quad (18)$$

where $\mathcal{I}(N)$ is the set of interior reserve managers, the constrained global equilibrium exists and is unique. The reported calibration satisfies this condition numerically over the full unit interval.

The global routine-use results are nonlinear scenarios. The local 2022 numbers remain exact comparative statics within the linear economy.

5.3 One freeze

Table 2 reports steady states, and Figure 1 the transition, for the single-use regime: the world learns in 2022 that the rule is $\bar{s} = 1$ with breadth $p = 0.02$.

In the single-use scenario, the aggregate dollar share declines 3.0 percentage points and captured privilege falls by \$1.7 billion a year. These are conditional model outputs. They do not estimate the causal effect of the 2022 freeze. At $m = 2.5$ the same wedge lowers the dollar share by 5.1 points. At $m = 1$ it lowers the share by 2.0 points. Figure 2 shows the aligned-country response implied by network feedback. That response is zero when $\ell_1 = 0$.

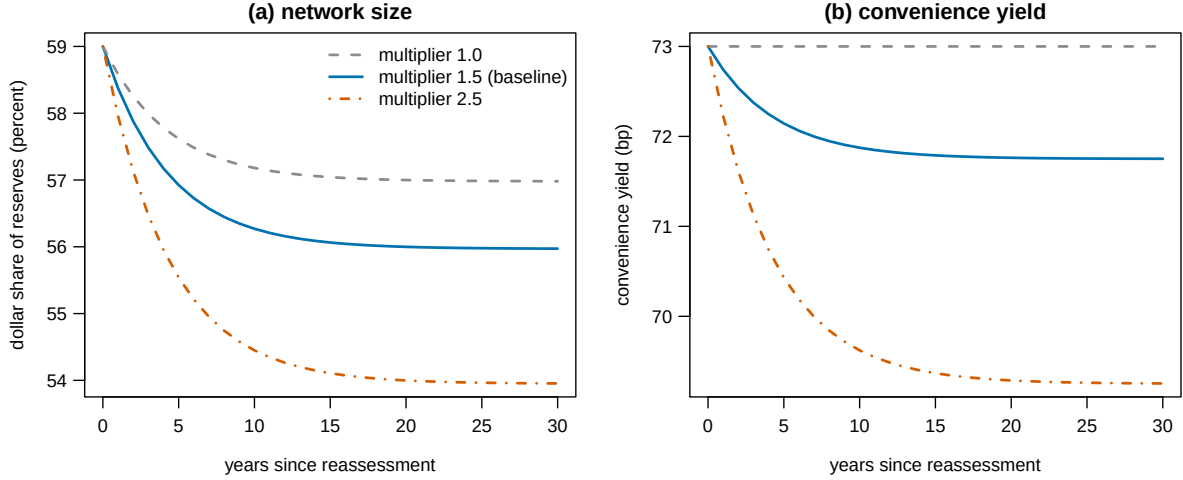


Figure 1: Transition after the 2022 reassessment (single-use regime, $\hat{\pi}$: $0 \rightarrow 5.0$ bp). Aggregate dollar share and convenience yield, for network multipliers 1.0, 1.5, 2.5.

5.4 The sanctions-use frontier

Figure 3 reports the sanctions-use frontier from the global economy. At the baseline inputs, full intensity and full restraint have the same flow value at $p^* = 5.5$ percent. This is a scenario crossing, not an estimate of reserve managers' beliefs. At $p = 25$ percent, the global equilibrium has $N = 23.2$ percent and captured privilege is lower by \$17.8 billion a year.

The threshold is the function $p^*(\gamma, \phi, \mu, m, \zeta_H, \chi)$. Holding other inputs at baseline, it falls from 10.7 to 2.7 percent as m rises from 1 to 2.5. It falls from 11.1 to 3.7 percent as ϕ rises from 0.25 to 0.75. Increasing χ from one to two lowers it from 5.5 to 2.8 percent. At $\chi = 4$ it is 1.4 percent. Changing ζ_H from 0.25 to one moves the crossing from 11.0 to 2.8 percent. With $\gamma = 800$ billion, full intensity remains above restraint throughout $p \in [0, 1]$, so no crossing exists on the feasible interval. The full reported scenario grid gives thresholds from 0.3 to 34.4 percent where a crossing exists.

The inverse threshold $\gamma^*(p, \phi, \mu, m, \zeta_H, \chi)$ is the geopolitical value required to justify full intensity. At baseline inputs it equals \$71.4 billion for $p = 0.02$, \$363.9 billion for $p = 0.10$, and \$710.0 billion for $p = 0.25$. This representation leaves γ as a policy input and states the model's tradeoff without assigning it an empirical value.

5.5 Rules against discretion

Figure 4 evaluates the local rule at alternative values of the policy input γ . At \$200 billion, $s_R = 0.83$ and the fixed-belief benchmark is one. The flow value of commitment is \$0.1 billion a year. At $\gamma = 50$ billion the rule is 0.30. At $\gamma = 800$ billion it is 0.96. These figures inherit the local approximation and $\zeta_H = 0.50$. They are not estimates of an optimal observed doctrine.

The quantitative exercise establishes thresholds conditional on reserve demand, network curvature, crisis liquidity, and issuer pass-through. Estimating those objects is therefore more

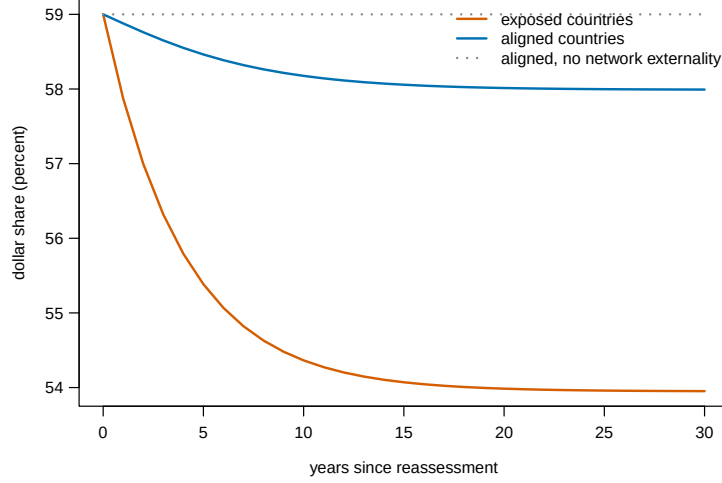


Figure 2: The network fingerprint: exposed and aligned dollar shares after the reassessment, baseline multiplier. The dotted path is the aligned share when $\ell_1 = 0$: with the network externality shut down, aligned countries stay fixed.

important than attaching confidence to a single baseline number.

6 Empirical Designs

The heterogeneous model turns on the distribution of direct responses and the network feedback in (7). No single regression identifies both. The empirical program therefore combines country holdings, aggregate reserve composition, and convenience-yield responses while treating custodial attribution as a measurement problem.

Design 1: country reserve demand. The main public outcome is foreign official holdings of United States Treasury securities reported by TIC, scaled by total reserves from the IMF International Financial Statistics. The event-study equation is

$$\Delta \left(\frac{\text{UST}_{i,t}}{\text{Reserves}_{i,t}} \right) = \alpha_i + \tau_t + \sum_{k \neq -1} \beta_k z_i \mathbf{1}\{t = t_0 + k\} + \Gamma' X_{i,t} + \varepsilon_{i,t}. \quad (19)$$

The exposure index is predetermined and continuous. It combines pre-2022 alignment, prior incidence of financial sanctions, reserve adequacy, external-debt currency, and trade-invoicing exposure. Directly sanctioned targets are excluded from the anticipation sample because their post-event holdings reflect seizure and payment restrictions rather than voluntary reserve demand. Country and time effects are supplemented by commodity terms, reserve accumulation, exchange-rate valuation, sovereign risk, and global safe-asset demand.

TIC reports the location of the foreign holder or custodian, not beneficial ownership in every case (United States Department of the Treasury, 2026). Custody centers are pooled into a separate group and omitted in a robustness sample. Official and private holdings are separated

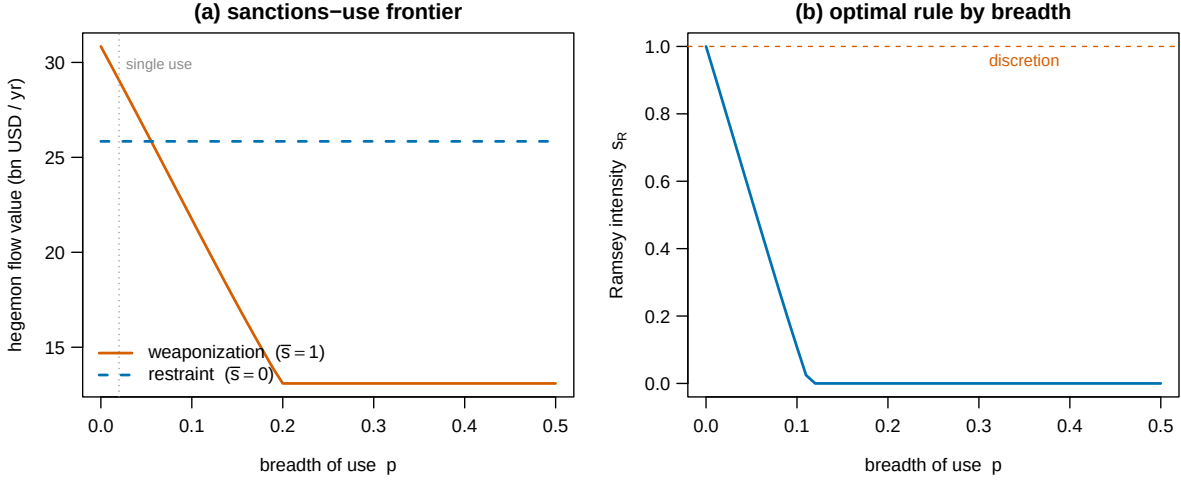


Figure 3: The sanctions-use frontier in the matched global economy. Panel (a) compares full intensity with restraint as anticipated breadth varies. Panel (b) reports the optimal commitment intensity.

when the published table permits. The disclosed central-bank subsample and gold purchases provide validation outcomes. COFER is not used at the country level because the IMF keeps individual submissions confidential (International Monetary Fund, 2026).

Design 2: aggregate network moments. Let Sal_t measure public announcements of asset freezes and payment restrictions in the Global Sanctions Data Base release 4 (Global Sanctions Data Base, 2025). The price equation is

$$CY_t = \rho CY_{t-1} + \delta Sal_t + \Xi' Z_t + u_t \quad (20)$$

where Z_t includes global risk, monetary policy, Treasury supply, dealer balance sheets, and exchange-rate hedging costs. The sign restriction is $\delta < 0$. This regression alone does not identify ℓ_1 because sanctions news can move global risk and expected Treasury supply. The multiplier is bounded jointly by the aggregate reserve response, aligned-country holdings, and dollar-specific convenience yields relative to other safe currencies. Variation across several episodes is required. The 2022 event by itself cannot separate a network response from concurrent global shocks.

Structural mapping. The country panel disciplines the distribution of $z_i \chi_i / \kappa_i$. Aggregate quantities and prices discipline the denominator in (7). Revision paths discipline adjustment costs only after the permanent component of the shock has been separated from transitory risk. The model is overidentified when the same parameters must fit exposed holdings, aligned holdings, aggregate COFER, gold accumulation, and convenience yields. The policy inputs γ and ζ_H remain outside that estimation.

Table 3 records restrictions that can reject the mechanism. A response to trade-only sanctions of the same size would reject the accessibility interpretation. A zero aligned response is consistent with $\ell_1 = 0$, though it can also arise from measurement error or offsetting common shocks. The network parameter is therefore inferred from the joint system rather than one coefficient.

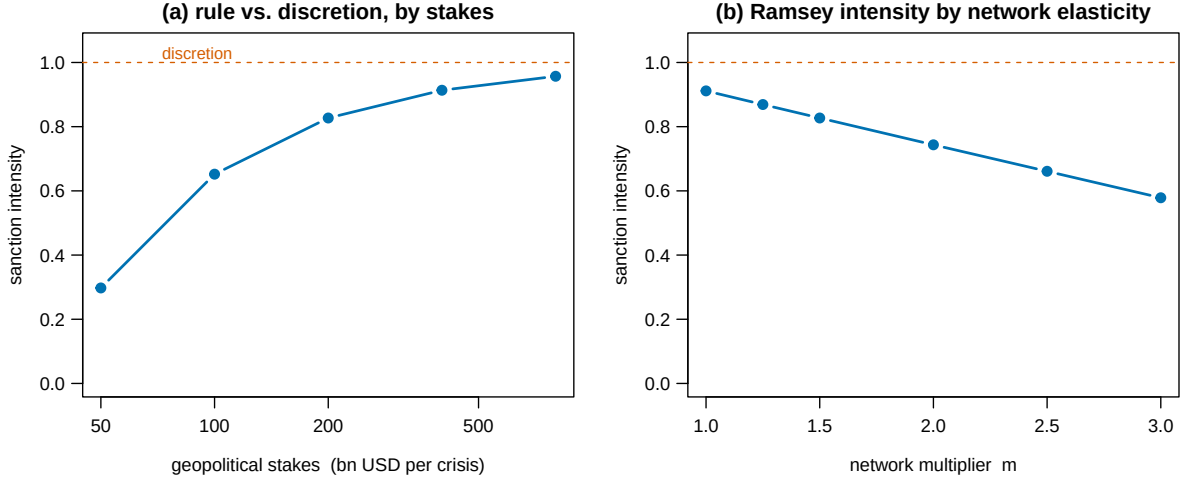


Figure 4: The weaponization bias. Panel (a): Ramsey intensity by geopolitical stakes γ (log scale). Discretion is 1 throughout. Panel (b): Ramsey intensity by network multiplier at $\gamma = 200$.

Gold purchases and aggregate COFER are validation moments (World Gold Council, 2024; Arslanalp et al., 2022). Neither identifies the causal effect of sanctions without additional variation. Table 4 records the public-data implementation.

7 Conclusion

Financial sanctions draw strength from the same reserve network that they can weaken. The paper closes that policy loop. In the local economy, accessibility risk enters reserve demand as a haircut on convenience and network complementarity amplifies the aggregate response. With heterogeneous countries, raw group differences no longer reveal a single portfolio curvature. The relevant object is the weighted response in (7).

The fixed-belief crisis benchmark chooses maximal intensity because current portfolios cannot react within the period. A public belief state changes that conclusion by placing future reserve demand in the current decision. Commitment and the public trigger strategy give transparent benchmark allocations. They do not establish how observed sanctions policy is formed.

The quantitative results are threshold functions. At the baseline scenario, full use and restraint cross at $p^* = 5.5$ percent. That number moves materially with crisis liquidity and issuer capture. The empirical content lies in the joint behavior of TIC holdings, aggregate COFER, gold purchases, and dollar convenience yields. Estimating those responses determines whether the relevant frontier lies near occasional use or far beyond it.

Table 3: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
Exposed reserve demand falls after accessibility reassessment	$-z_i \omega p_i \phi_i \chi_i / \kappa_i$ locally	0	TIC and disclosures
Response rises with predetermined exposure	dose response	flat	event study
Asset freezes exceed trade-only sanctions	larger absolute response	equal	GSDB split
Dollar convenience falls with financial-sanctions salience	negative cumulative response	0	yield series
Aligned holdings move with the network	sign of $\ell_1 \Delta N$	0 if $\ell_1 = 0$	reserve panel
Permanent and transitory components differ	persistent component	full reversion	event path

Notes: predictions are local sign and mapping restrictions. TIC measures custodial location imperfectly. COFER supplies only aggregate currency composition.

Table 4: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or proxied	Measure
IMF COFER	aggregate shares, 1999–	quarterly	public	observed (aggregate only)	in hand
TIC official holdings	by country, monthly, custodial bias	monthly	public	proxy for beneficial ownership	in hand
Disclosed CB composition	subsample of central banks	annual	public reports	observed, selected sample	assembled from reports
WGC gold purchases	by country	quarterly	public	observed	in hand
GSDB release 4	1950–2023	episode	public academic	observed	in hand
UN ideal points	all members, 1946–	annual	public	observed	in hand
Convenience-yield series	US, G10	monthly	replicable	observed	in hand

Notes: the salience index uses public GSDB events. Disclosed reserve composition is selected because publication is voluntary. TIC country attribution is custodial rather than complete beneficial ownership.

A Proofs

A.1 Proof of Proposition 1

In the homogeneous benchmark, $\kappa > 0$ makes (3) strictly concave in a . The interior first-order condition gives (4). Holding N fixed gives $\partial a_i / \partial \hat{\pi} = -z_i / \kappa$. Subtracting the exposed policy from the aligned policy removes the common network term and gives $a_A - a_E = \hat{\pi} / \kappa$. The exposed first-order condition can be written $\kappa a_E = \bar{r} + [\ell(N) - \hat{\pi}]$, which proves the convenience-yield statement. ■

A.2 Proof of Proposition 2

Define the aggregate best response $B(N) = [\bar{r} + \ell_0 + \ell_1 N - \mu \hat{\pi}] / \kappa$. On the interior region its Lipschitz constant is ℓ_1 / κ . If $\ell_1 < \kappa$, the contraction theorem gives a unique fixed point. Solving it gives (5). Total differentiation yields $dN / d\hat{\pi} = -(\mu / \kappa) / (1 - \ell_1 / \kappa)$. When $\ell_1 \geq \kappa$, this proof supplies no uniqueness conclusion. The constrained fixed-point problem remains continuous and therefore has an equilibrium, but its number and location depend on intercepts and binding portfolio bounds. ■

A.3 Proof of Corollary 1

The cross-sectional equality follows from Proposition 1. Proposition 2 gives the aggregate response. Their ratio is m in the homogeneous interior economy. Recovering that ratio requires both moments. ■

A.4 Proof of Proposition 3

Strict concavity gives the unique interior policy (6). Differentiating it with respect to $\bar{s}$ and integrating gives

$$\frac{dN}{d\bar{s}} = \ell_1 A_\lambda \frac{dN}{d\bar{s}} - A_z.$$

The maintained inequality makes the denominator positive and gives (7). For country i , the derivative holding N fixed is $-z_i \omega p_i \phi_i \chi_i / \kappa_i$. The total derivative adds $(\lambda_i \ell_1 / \kappa_i) dN / d\bar{s}$. Hence a group contrast removes the network term only when the relevant distributions of λ_i / κ_i coincide. ■

A.5 Proof of Proposition 4

Averaging (8) over countries and collecting terms gives

$$\beta_c \nu N_{t+1} - (\kappa + (1 + \beta_c) \nu - \ell_1) N_t + \nu N_{t-1} = -(\bar{r} + \ell_0 - \mu \hat{\pi}),$$

and subtracting the aligned from the exposed equation gives the same recursion for d_t with $\ell_1 = 0$ and forcing $\hat{\pi}$ (the network term is common and cancels). Let $P(\lambda) = \beta_c \nu \lambda^2 - B\lambda + \nu$

with $B = \kappa + (1 + \beta_c)\nu - \ell_1$. Then $P(0) = \nu > 0$ and $P(1) = \beta_c\nu - B + \nu = \ell_1 - \kappa < 0$, so one real root lies in $(0, 1)$. The product of the roots is $1/\beta_c > 1$, and their sum is $B/(\beta_c\nu) > 0$, so the second root is positive and exceeds one. There is exactly one stable root $\lambda \in (0, 1)$, and the unique bounded solution given the predetermined initial condition is the stated first-order recursion, with the constant equal to the new steady state. The stable root

$$\lambda = \frac{B - \sqrt{B^2 - 4\beta_c\nu^2}}{2\beta_c\nu}$$

is decreasing in B . The implicit derivative gives the same sign: at the stable root the polynomial crosses from positive to negative, so $P_\lambda(\lambda) < 0$ and $d\lambda/dB = \lambda/P_\lambda < 0$. Since B is decreasing in ℓ_1 , $\lambda_N > \lambda_d$ and $\partial\lambda_N/\partial\ell_1 > 0$. ■

A.6 Proof of Proposition 5

Under fixed beliefs, $G'(s) = \gamma(1 - s)$ and $s_D = 1$. For commitment, substitute $N(\bar{s}) = \bar{N}_0 - \omega c\bar{s}$, where $c = \mu p\phi\chi/(\kappa - \ell_1)$, into (11). The second derivative is $2\zeta_H\ell_1\omega^2c^2W - \omega\gamma$, which is negative under the stated condition. The first-order condition uses $\Omega'(N) = \zeta_HW(\ell_0 + 2\ell_1N)$ and solves to (12). Rearrangement gives

$$1 - s_R = \frac{\zeta_HcW[\ell_0 + 2\ell_1N(1)]}{\gamma - 2\zeta_H\ell_1\omega^2c^2W} > 0.$$

The inequality uses the positive marginal captured privilege at $N(1)$. The value of commitment is the quadratic vertex gap

$$V(s_R) - V(1) = \frac{1}{2}(\omega\gamma - 2\zeta_H\ell_1\omega^2c^2W)(1 - s_R)^2 > 0. \quad \blacksquare$$

A.7 Proof of Proposition 6

Sustainability of the trigger profile requires the one-time deviation gain to be bounded above by the discounted value of conformity:

$$\frac{\beta_H}{1 - \beta_H} [V(s_R) - V(1)] \geq G(1) - G(s_R) = \frac{\gamma}{2}(1 - s_R)^2.$$

The vertex formula in the proof of Proposition 5 gives $V(s_R) - V(1) = \frac{1}{2}(\omega\gamma - 2\zeta_H\ell_1\omega^2c^2W)(1 - s_R)^2$. Canceling the common factor $(1 - s_R)^2/2$ yields (14). The interior concavity condition makes its denominator positive. ■

A.8 Proof of Proposition 7

For any N , strict convexity of C_i gives a unique constrained reserve choice. Projection onto $[0, 1]$ is continuous, so the aggregate best response maps the unit interval into itself and has a fixed point. At an interior choice, its derivative with respect to N is $\lambda_i\ell^{g'}(N)/C_i''(a_i)$. It is zero at

a strict corner. Integrating gives the expression inside braces in (18). The strict upper bound makes the aggregate map a contraction and proves uniqueness. ■

B The Saddle Path and the ν Calibration

The characteristic quadratic of Proposition 4 can be inverted for the adjustment cost that delivers a target persistence: setting $P(\lambda) = 0$ and solving for ν ,

$$\nu = \frac{(\kappa - \ell_1) \lambda}{(1 - \lambda)(1 - \beta_c \lambda)},$$

so a three-year aggregate half-life ($\lambda = 2^{-1/3}$) pins $\nu = 0.133$ at the baseline (κ, ℓ_1) . The full transition in levels is recovered from the pair (N_t, d_t) by $a_{A,t} = N_t - \mu d_t$ and $a_{E,t} = a_{A,t} + d_t$. Because the two recursions are driven by different roots, the model makes a joint prediction about timing: the exposed-aligned gap (root λ_d , half-life 2.4 years) closes on its plateau faster than the aggregate share (root λ_N , half-life 3 years), which Design 1 and the COFER aggregate can check against each other.

C Corners and Tipping

With common κ and $\ell_1 < \kappa$, projection onto $[0, 1]$ cannot increase the slope of the local aggregate best response. The constrained local equilibrium is therefore unique. This result does not govern the routine-use calculation, which uses (15) and (16). Proposition 7 supplies its uniqueness condition. If the contraction inequality fails, the model permits multiple fixed points but does not require them. Tipping is then a possible global outcome whose relevance must be established from the estimated slope of the aggregate best response.

D Parameter Discipline and Sensitivity

Table 5 records the status and range of each object. At $\ell_1 = 0$, aligned portfolios do not respond through the network. When $\omega p \phi \chi$ approaches zero, the accessibility wedge vanishes. As γ grows, the commitment and fixed-belief allocations converge.

E Data Appendix

Reserve composition. IMF COFER reports quarterly aggregate currency shares of allocated reserves (International Monetary Fund, 2026). Individual-country reports are confidential. The aggregate dollar series is valuation-adjusted before it is used as a quantity moment (Arslanalp et al., 2022).

Table 5: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range exam- ined	Effect of the range
$\ell(N_0) = 73$ bp	eq. (1)	bp per year	external estimate	observed un- certainty	privilege scales with yield
$N_0 = 0.59$	eq. (5)	share	aggregate COFER	observed un- certainty	pins κ with $\ell(N_0)$
$\kappa = 124$ bp	eq. (4)	bp per share	implied, $\kappa = \ell(N_0)/N_0$	-30% (outside- asset case)	erosion 3.0 $\rightarrow$ 5.5 pp
$m(\ell_1) = 1.5$	eq. (5)	unitless	scenario, joint mo- ments	1 to 2.5	erosion 2.0–5.1 pp
$\nu = 0.133$	eq. (8)	utility	scenario half-life	timing alter- natives	steady states unaf- fected
$\beta_c = 0.96$	eq. (8)	annual	externally fixed	annual dis- count	negligible near 1
$\mu = 0.5$	eq. (5)	reserve share	scenario	0.3 to 0.7	threshold 9.1–4.0%
$\omega = 0.05$	eq. (2)	per year	scenario	alternative ar- rival rates	wedge and G scale
$\phi = 0.5$	eq. (2)	share frozen	observed episode	0.25 to 0.75	threshold 11.1–3.7%
$p = 0.02$	eq. (2)	probability	scenario	0 to 1	sanctions-use frontier
$\chi = 1$	eq. (2)	ratio	scenario lower bound	1 to 4	threshold 5.5–1.4%
$\zeta_H = 0.50$	eq. (9)	share	scenario	0.25 to 1	threshold 11.0–2.8%
$W = \$12$ tn	eq. (9)	USD	aggregate COFER	observed un- certainty	dollar magnitudes scale
$\gamma = \$200$ bn	eq. (10)	USD per crisis	policy input	50 to 800	s_R 0.30–0.96

Notes: only the product $\omega\bar{s}p\phi\chi$ enters homogeneous reserve demand. Policy thresholds depend separately on the arrival rate because geopolitical benefits are realized in crisis states.

Country-level holdings. TIC reports monthly foreign official holdings by reported country (United States Department of the Treasury, 2026). This is custodial attribution and need not equal beneficial ownership. Custody hubs are aggregated and then omitted in a robustness sample. Country reserve totals come from IMF International Financial Statistics. Gold purchases come from World Gold Council releases (World Gold Council, 2024).

Sanctions. Global Sanctions Data Base release 4 covers episodes through 2023 (Global Sanctions Data Base, 2025). The salience index retains asset freezes and payment restrictions. Trade-only measures form a placebo group. The 2022 freeze magnitude is documented separately (Ribakova et al., 2025).

Alignment. Ideal-point distances from Bailey et al. (2017), updated through the sample. The exposure measure is the pre-2022 average distance from the G7 mean, so that exposure is predetermined at the event.

Convenience yields. The Krishnamurthy and Vissing-Jorgensen (2012) AAA–Treasury spread, the Du et al. (2018) Treasury premium (five-year, G10 average and by currency), and the Jiang et al. (2021) Treasury basis. Global controls: VIX, term premia, and Federal Reserve balance-sheet variables.

F Extensions

Endogenous outside asset. Let the outside asset’s issuer invest to raise its own network quality, lowering κ over time. The outside-asset counterfactual of Section 5 is the comparative static: each step down in κ raises the erosion per use and lowers the Ramsey intensity. A full dynamic treatment makes κ_t a state in the hegemon’s problem and turns the model into a duopoly of networks, the race sketched by Farhi and Maggiori (2018).

The target’s margin. Freezing a sovereign’s reserves degrades its crisis insurance and can push it toward default, imposing credit losses on the hegemon’s own financial system. Bianchi and Sosa-Padilla (2024) model the margin. In the present notation those losses subtract from G , reducing γ and, by Proposition 5, widening the gap between discretion and any rule a planner would choose.

Continuous exposure. Proposition 3 already permits $z_i \in [0, 1]$. Its direct response is proportional to $z_i \omega p_i \phi_i \chi_i / \kappa_i$. A monotone raw dose response requires the remaining components to be held fixed or modeled conditionally.

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